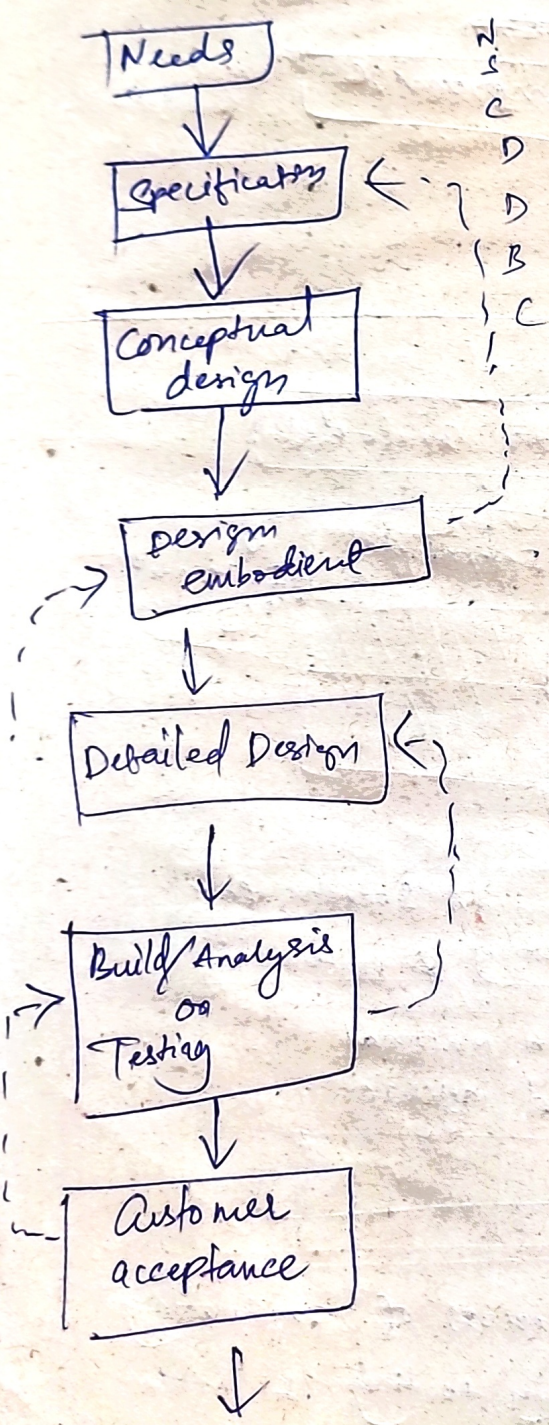


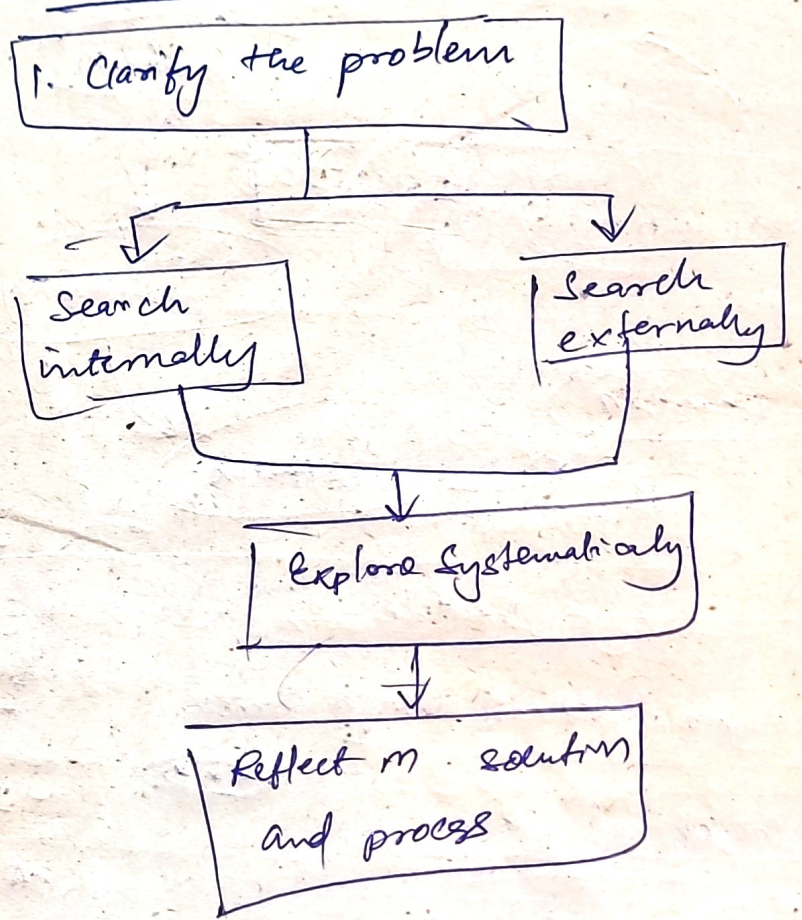
Mod - 1, 2 :

2-step-model :

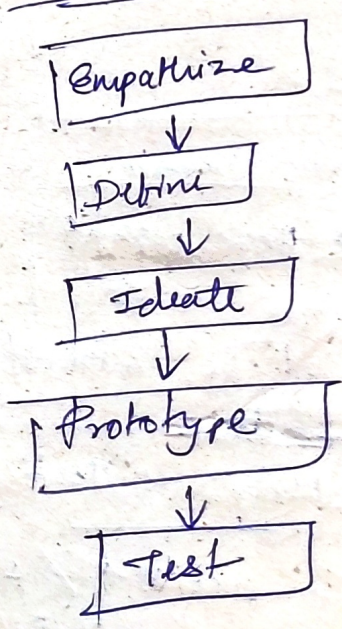
A motivation to design



Concept Generation Process :



Design Thinking Process :



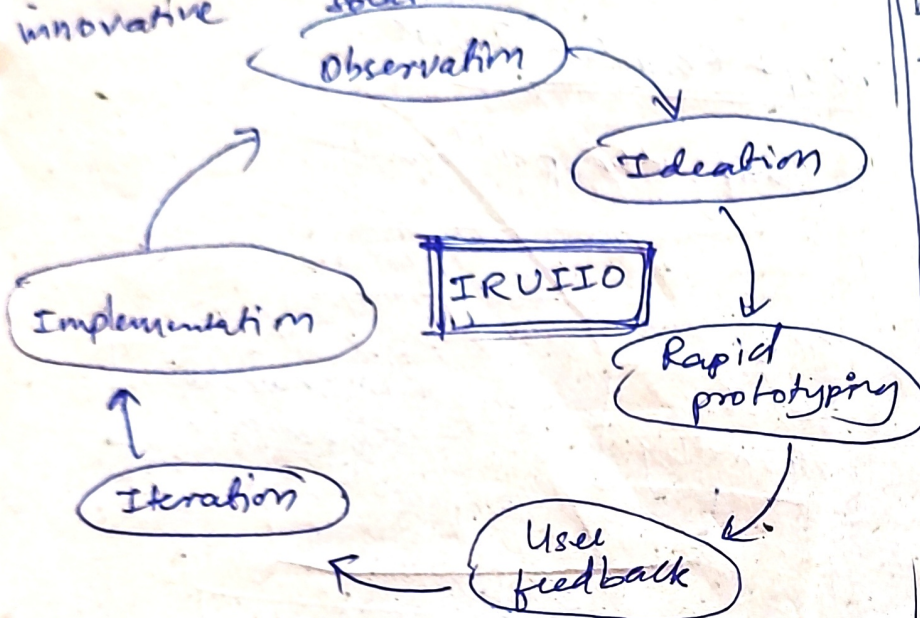
EDLPT

Empathy Map :

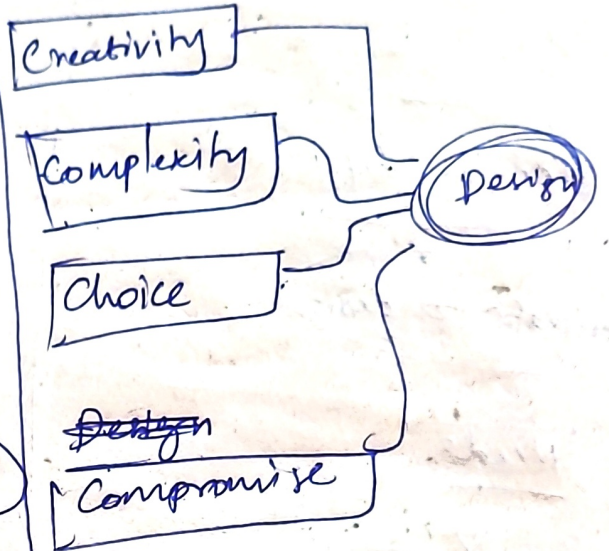
says	Thinks
does	Feels

→ Human Centred Design: - (HCD)

• Starts with people ends with innovative solutions.



→ Four C's of design:



Module - 2

→ Major phases of Design Process:

→ Conceptual Design

→ Embodiment Design

→ Detailed Design

Aesthetic Design

→ Vision

→ Hearing

→ Touch

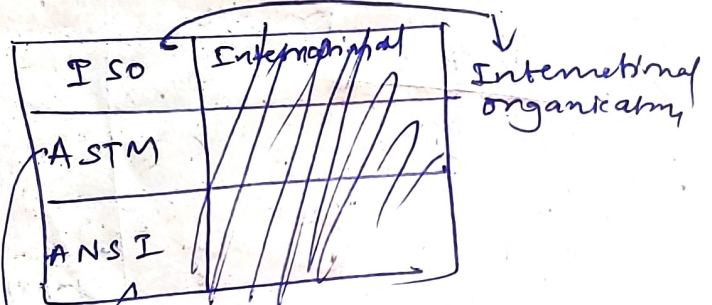
→ Taste & smell

Optimum Design

• Best possible design that could be constructed.

Usability Testing: A process of evaluating a product/service by testing it with representative users.

Standards & codes



American Society for Testing & materials

American national standard institute

Module - 3

→ Sustainable Development

→ Social sustainability → Human rights

fair labor practices, living conditions,
affordable housing, physical &
mental medical support, employment
opportunities

→ Environmental sustainability

→ Economic sustainability

⇒ Lifecycle Assessment: [LCA]

A systematic, standardized
approach to quantifying the
potential environmental impacts of
a product or process that
occur from raw material
extraction to end of life

⇒ Biomimicry

⇒ Eco-design

⇒ Social Innovation

Agnisumukh

* Ultra-efficient
cooking stones
saves 30% on gas.

* Improves cooking.

* Helps to beat
indoor pollution in
commercial kitchen.

Engineering materials :-

* Criteria for selecting an
Engineering material.

→ Properties of Material

→ Cost of Material

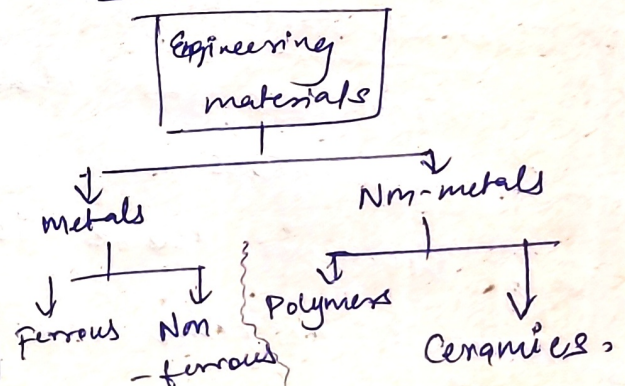
→ Availability of Material

→ Service in life of material.

→ Appearance of material.

Substances that
are useful in the
field of engineering

Classification:



⇒ Stress & Strain :-

$$\text{Stress} = \frac{\text{load}}{\text{area}}$$

(Elasticity)

No-unit

$$\text{Strain} = \frac{\text{change in length}}{\text{original length}}$$

(plasticity)

$$\text{Stress } (\sigma) = \frac{\text{Force } (F)}{\text{Area } (A)}$$

$$= \frac{N}{m^2}$$

→ Tensile stress - Pulling apart

→ Compressive stress - Pushing together

→ Shear stress - Forces acting parallel to the surface

Strain: (ϵ)

$$\epsilon = \frac{\text{change in length } (\Delta L)}{\text{original length } (L)}$$

→ Longitudinal strain - Due to axial loading -

→ Lateral strain - Perpendicular to axial load.

→ Volumetric strain - change in volume.

⇒ Hooke's Law :-

Stress $\propto$ Strain

→ proportionality constant

$$\sigma = E \epsilon$$

⇒ Thermal Stresses :

* Stress due to change in temperature.

* Whenever there is some increase/decrease in the temperature of a body, it causes the body to expand/shrink.

PPT into slide

With 46 pages

file name: "Engineering materials"

Module - 4

Metaphor method

Concept-generation

using one entity like swan to represent another (eg. chair designed from the idea of swan).

→ Types

- Descriptive - "Chair looks like swan".
- Generative - "Design a chair like a swan".

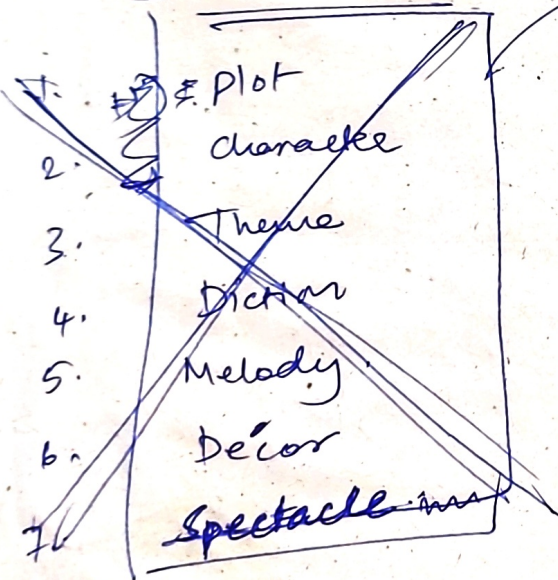
→ Higher-order concept generation

→ Designing something, that is beyond existing categories.

→ Creativity

→ Problem-solving.

→ Storytelling in Design:



Sketching

* Quick, informal, draft to explore ideas.

Mediums: Pen, pencil, any other ways.

* Helps in early design decisions, cheap and editable.

7-elements:

- * Plot
- * Character
- * Theme
- * Dictum
- * Melody
- * Décor
- * Spectacle

Dynamic Diagram

→ Visuals showing movement or interaction in a system.

→ used to demonstrate process flow/system behaviour.

~~10~~ ⇒ X-scripts : (User Experience Scripting)

• text-based tool to describe user interaction with a system.